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END SEMESTER EXAMINATION - 2025

Name of the Course: B. Tech

Name of the Paper: **Industrial Engineering & Project Management**

Time: 3 Hours

Semester: VI

Paper Code: TME 609

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(20marks)	CO 1
(a)	Define Production Planning and Control. Explain its role in different types of production systems.	
(b)	Explain routing, scheduling, and dispatching with an example of batch production.	
(c)	A company has fixed costs of ₹1,00,000. The selling price per unit is ₹200 and variable cost per unit is ₹150. Find the break-even point.	
Q2	(20marks)	CO 2
(a)	Explain the EOQ model with assumptions.	
(b)	What is ABC analysis? How does it help in inventory control?	
(c)	Calculate EOQ for an annual demand of 5000 units, ordering cost ₹200, and holding cost ₹50.	
Q3	(20marks)	CO 3
(a)	Define productivity. Suggest ways to improve it.	
(b)	Explain symbols used in method study and draw a man-machine chart.	
(c)	Differentiate between process and product layout with neat diagrams.	
Q4	(20marks)	CO 4
(a)	Define Gantt chart and explain its advantages.	
(b)	What is a network diagram? Draw a sample network using CPM for 5 activities.	
(c)	Find expected time and variance for Activity A with optimistic=2, most likely=4, pessimistic=6.	
Q5	(20marks)	CO 5
(a)	What are the major project risks and how do you manage them?	
(b)	Discuss the stages of team formation in project management.	
(c)	Define EIA. Why is it important in project planning?	

